Unequal Impacts of School Desegregation Policies: Barcelona's Shock Plan

Preliminary findings | Please do not cite.

Desegpol
School desegregation policies in the city of Barcelona

GEPS
Globalisation, Education & Social Policies

UAB
Universitat Autònoma de Barcelona

School desegregation policies

- Policy reforms to tackle school segregation focused on two main areas (McPherson, 2011; Cummings et al., 2023; Grenet et al, 2021):
  - **School choice and admissions**: controlled school choice systems, priority access or quotas for socially disadvantaged students.
  - **School funding**: needs-based or formula funding schemes.

- **Limited short-term impact on school segregation and integration** (Danhier & Friant, 2019; Honey & Carrasco, 2023; Elacqua & Kutscher, 2023)

Barcelona's Shock Plan

Policy context

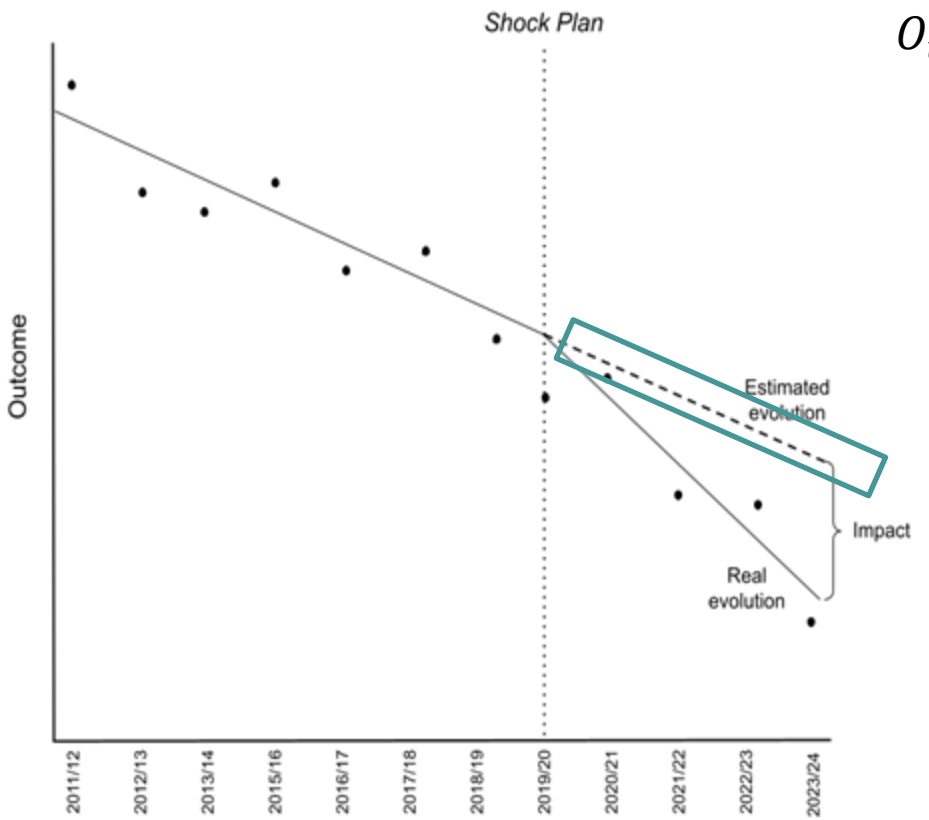
- **Important role of private subsidised schools**: 50.3% of enrolment in pre-primary and primary education, and 59.2% in lower secondary education.
- **Controlled school choice system** (catchment areas, quotas, centralised admission system), **but high families' capacity of choice** → 18.5 schools on average (Bonal et al., 2021).

Shock Plan Against School Segregation (SP)

- Improvement of **low-SES students targeting**
- **Preassigned school seats** for low-SES students (eliminated in 2021/2022)
- **Priority access quotas** for low-SES students, equal to the percentage of the low-SES students residing in the catchment area.
- **Additional funding** for low-SES students to cover the schooling cost (add-on tuition fees, material, extra-curricular activities, etc).

Methodological Approach

(Short) Interrupted Time Series



$$O_t = \beta_0 + \beta_1 T_t + \beta_2 SP_t + \beta_3 TSP_t + \mu_t$$

O_t School segregation/concentration indicator

T_t Time/school year (1,2,3…)

SP_t Intervention (0/1)

TSP_t Time after the intervention

Impact measurement

β_1 Pre-intervention trend

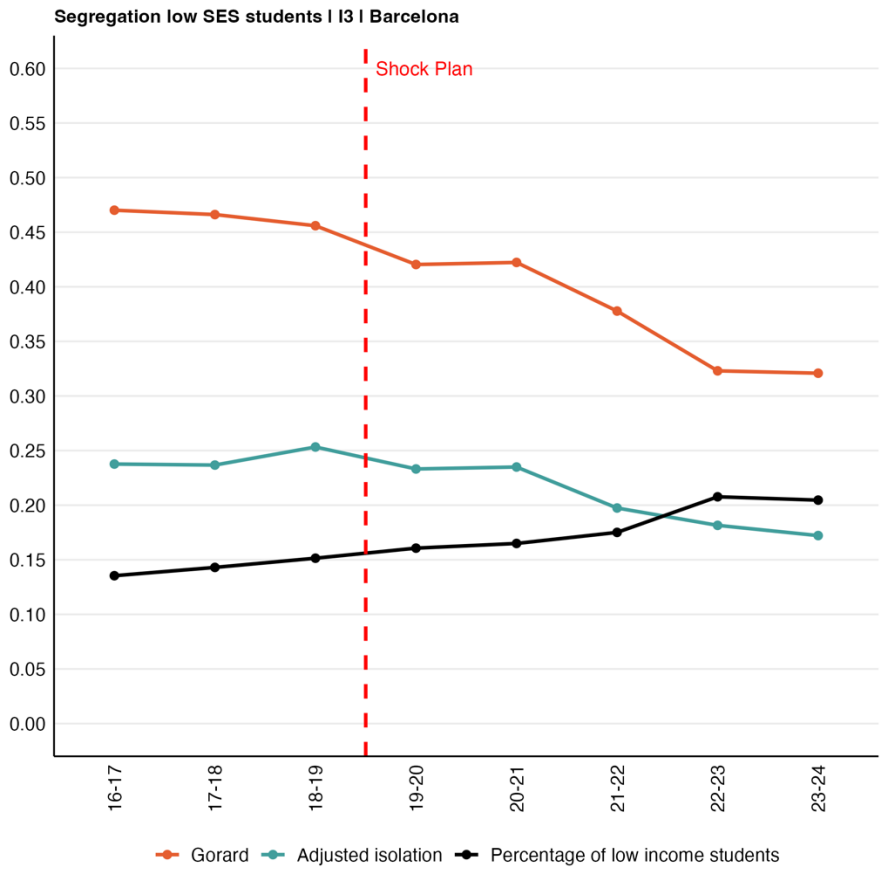
β_2 Immediate change after the intervention

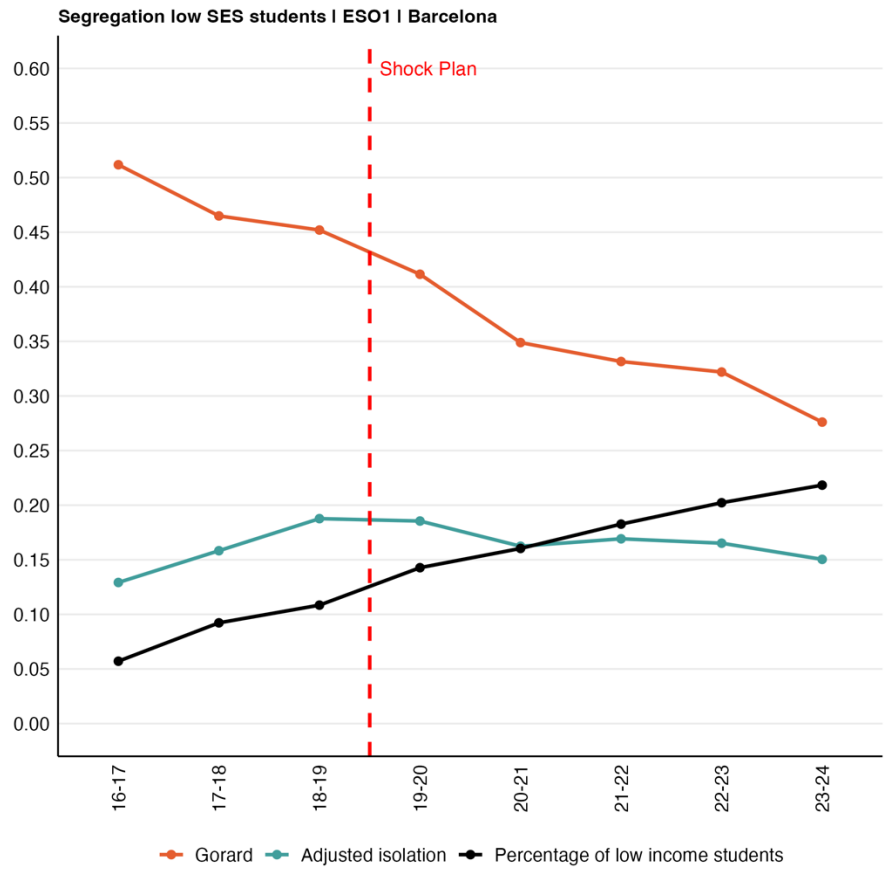
β_3 Post-intervention trend

Research design and operativisation

- **Two main indicators of impact (O_t):**
  - Gorard index (segregation)
  - Adjusted Isolation Index (concentration).
- **Pre-intervention** (2016/17-2018/19) vs. **Post-intervention** (2019/20-2023/24).
- **Choice moments**:
  - 1st grade of pre-primary education (I3) (N=13.000/year)
  - 1st grade of lower secondary education (ESO1) (N=14.500/year)
- **Focus on the target population** (≠ beneficiaries): low-SES students
  - Means-tested approach in the implementation process, most targeting based on diagnostics from educational or social services.
  - Imputation process considering the students classified as low-SES students after school choice.

The Evolution of Segregation of Low-SES students





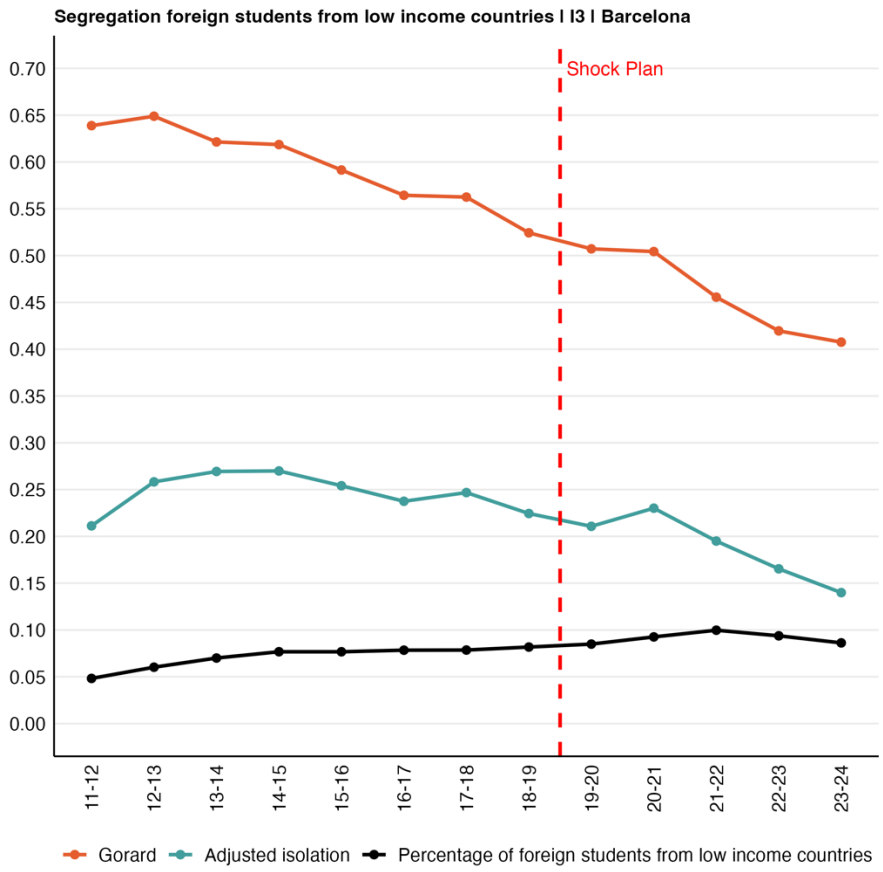
Impact of the Shock Plan on the segregation of Low-SES students

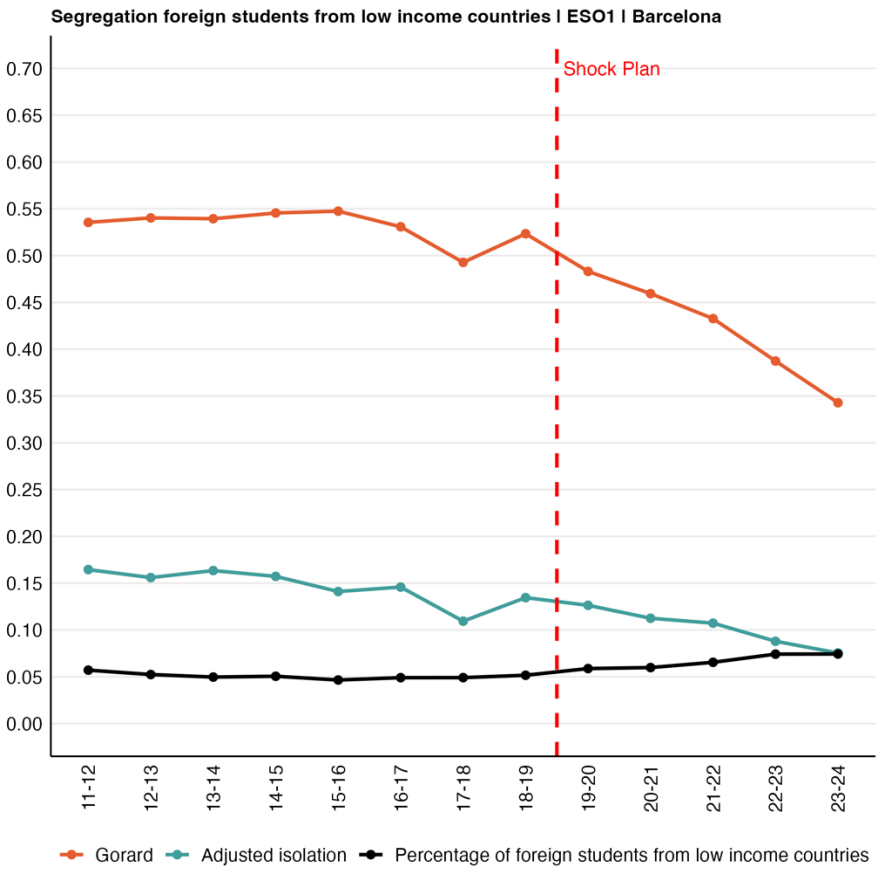
Table 1: **Low-SES students**

	I3		*ESO1*	
	Gorard	Adjusted Isolation	Gorard	Adjusted Isolation
	(1)	(2)	(3)	(4)
T (Pre-intervention trend)	−0.007	0.008	−0.030*	0.029***
	(0.011)	(0.007)	(0.011)	(0.005)
SP (Immediate impact)	0.005	0.006	−0.019	−0.001
	(0.022)	(0.013)	(0.022)	(0.010)
T-SP (Post-intervention trend)	−0.023	−0.025**	0.0001	−0.036***
	(0.012)	(0.007)	(0.012)	(0.005)
Constant	0.478***	0.227***	0.536***	0.100***
	(0.024)	(0.014)	(0.024)	(0.011)
Observations	8	8	8	8
R^2	0.960	0.946	0.979	0.923
Adjusted R^2	0.930	0.905	0.964	0.865
Residual Std. Error (df = 4)	0.016	0.009	0.016	0.007
F Statistic (df = 3; 4)	32.096***	23.309***	63.440***	15.964**

Note: *p<0.1; **p<0.05; ***p<0.01

Evolution | Foreign students low-income countries





Impact | Foreign students low-income countries

Table 2: **Foreign students from low-income countries**

	I3		*ESO1*	
	Gorard	Adjusted Isolation	Gorard	Adjusted Isolation
	(1)	(2)	(3)	(4)
T (Pre-intervention trend)	-0.017^{***}	-0.001	-0.004^{*}	-0.006^{***}
	(0.002)	(0.003)	(0.002)	(0.001)
PX (Immediate impact)	0.007	0.007	0.009	0.015
	(0.015)	(0.026)	(0.017)	(0.012)
TPX (Post-intervention trend)	-0.011^{**}	-0.020^{**}	-0.031^{***}	-0.007^{*}
	(0.004)	(0.007)	(0.005)	(0.003)
Constant	0.673^{***}	0.251^{***}	0.550^{***}	0.174^{***}
	(0.010)	(0.016)	(0.011)	(0.008)
Observations	13	13	13	13
R^2	0.983	0.789	0.967	0.916
Adjusted R^2	0.977	0.718	0.956	0.889
Residual Std. Error (df = 9)	0.012	0.021	0.014	0.010
F Statistic (df = 3; 9)	169.538^{***}	11.190^{***}	87.844^{***}	32.919^{***}

Note: *p<0.1; **p<0.05; ***p<0.01

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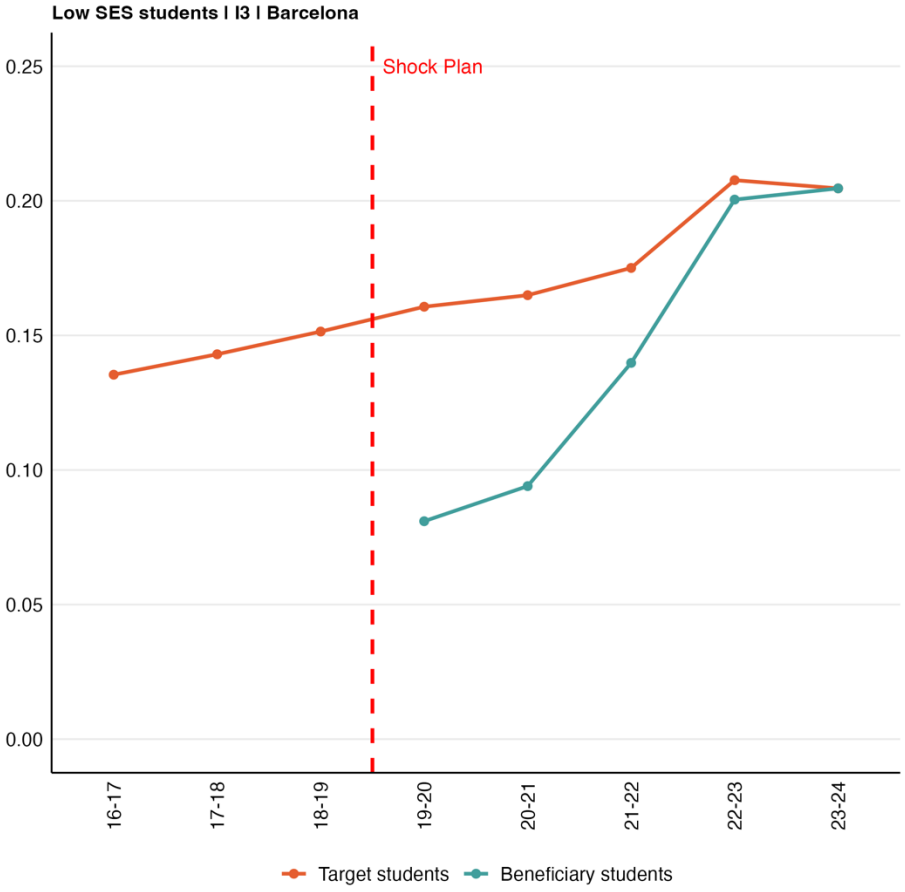
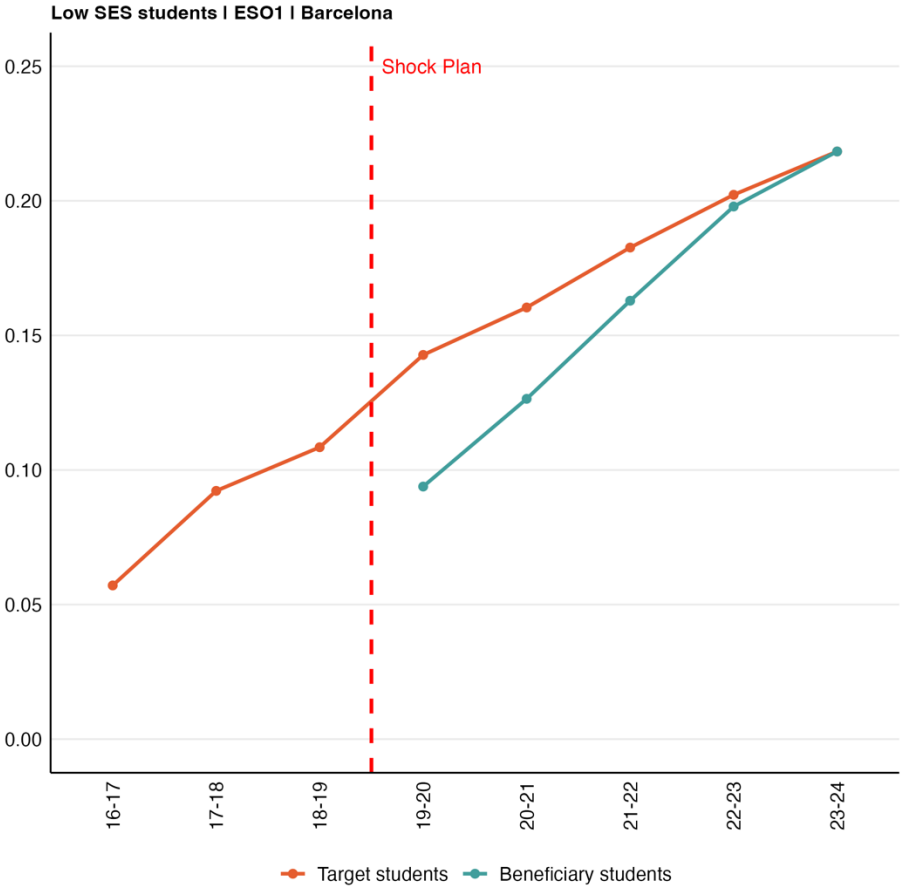
Understanding the (lack of) impact: a question of targeting?





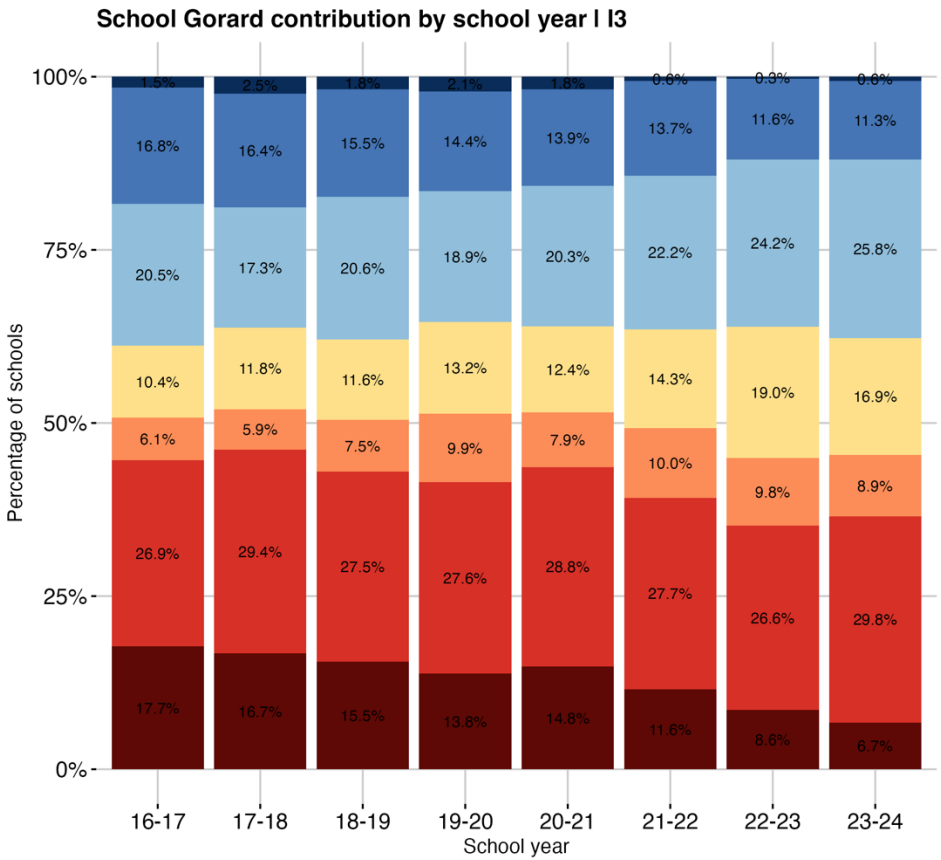
Understanding the (lack of) impact: a question of targeting?

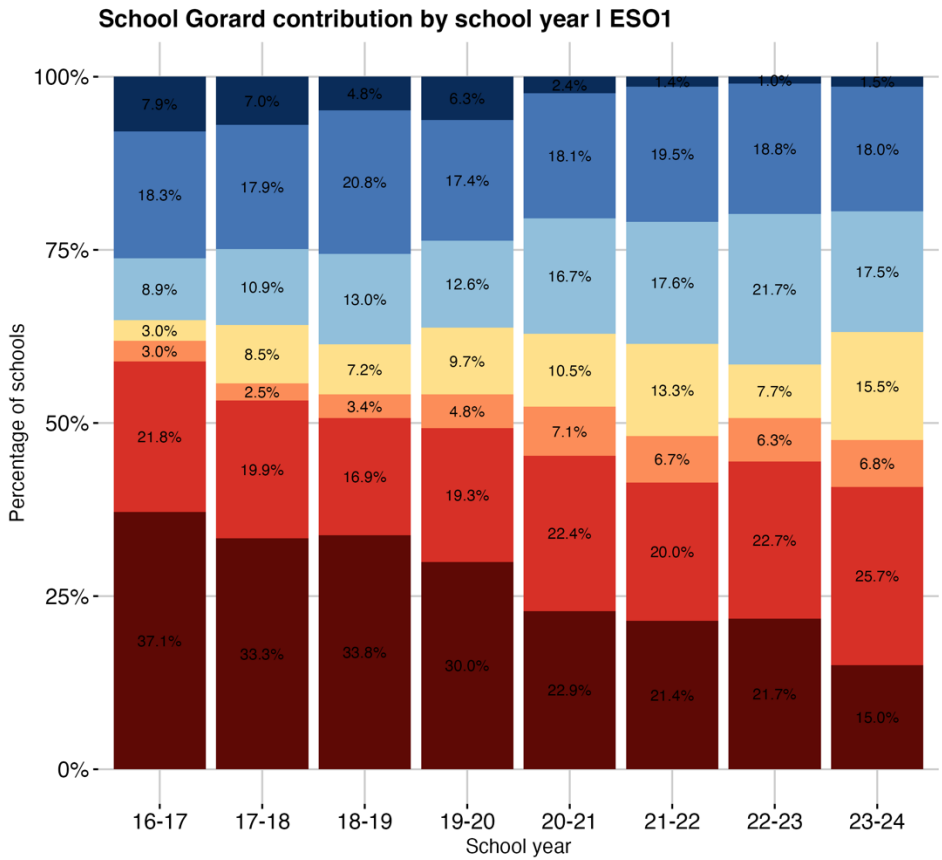
Table 3: Target and beneficiaries — Low-SES students

	I3				*ESO1*			
	Target		*Beneficiaries*		*Target*		*Beneficiaries*	
	Gorard	Adjusted Isolation	Gorard	Adjusted Isolation	Gorard	Adjusted Isolation	Gorard	Adjusted Isolation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T (Pre-intervention trend)	−0.007	0.008	−0.132***	0.015	−0.030*	0.029***	−0.078**	0.027***
	(0.011)	(0.007)	(0.018)	(0.008)	(0.011)	(0.005)	(0.022)	(0.005)
SP (Immediate impact)	0.005	0.006	−0.093*	−0.024	−0.019	−0.001	−0.125**	−0.037**
	(0.022)	(0.013)	(0.035)	(0.016)	(0.022)	(0.010)	(0.043)	(0.011)
T-SP (Post-intervention trend)	−0.023	−0.025**	0.098***	−0.004	0.0001	−0.036***	0.046	−0.018**
	(0.012)	(0.007)	(0.019)	(0.009)	(0.012)	(0.005)	(0.024)	(0.006)
Constant	0.478***	0.227***	0.973***	0.100***	0.536***	0.100***	0.794***	0.073***
	(0.024)	(0.014)	(0.038)	(0.018)	(0.024)	(0.011)	(0.047)	(0.012)
Observations	8	8	8	8	8	8	8	8
R^2	0.960	0.946	0.990	0.847	0.979	0.923	0.980	0.919
Adjusted R^2	0.930	0.905	0.983	0.731	0.964	0.865	0.965	0.858
Residual Std. Error (df = 4)	0.016	0.009	0.025	0.011	0.016	0.007	0.031	0.008
F Statistic (df = 3; 4)	32.096***	23.309***	133.996***	7.357**	63.440***	15.964**	65.860***	15.085**

Note: *p<0.1; **p<0.05; ***p<0.01

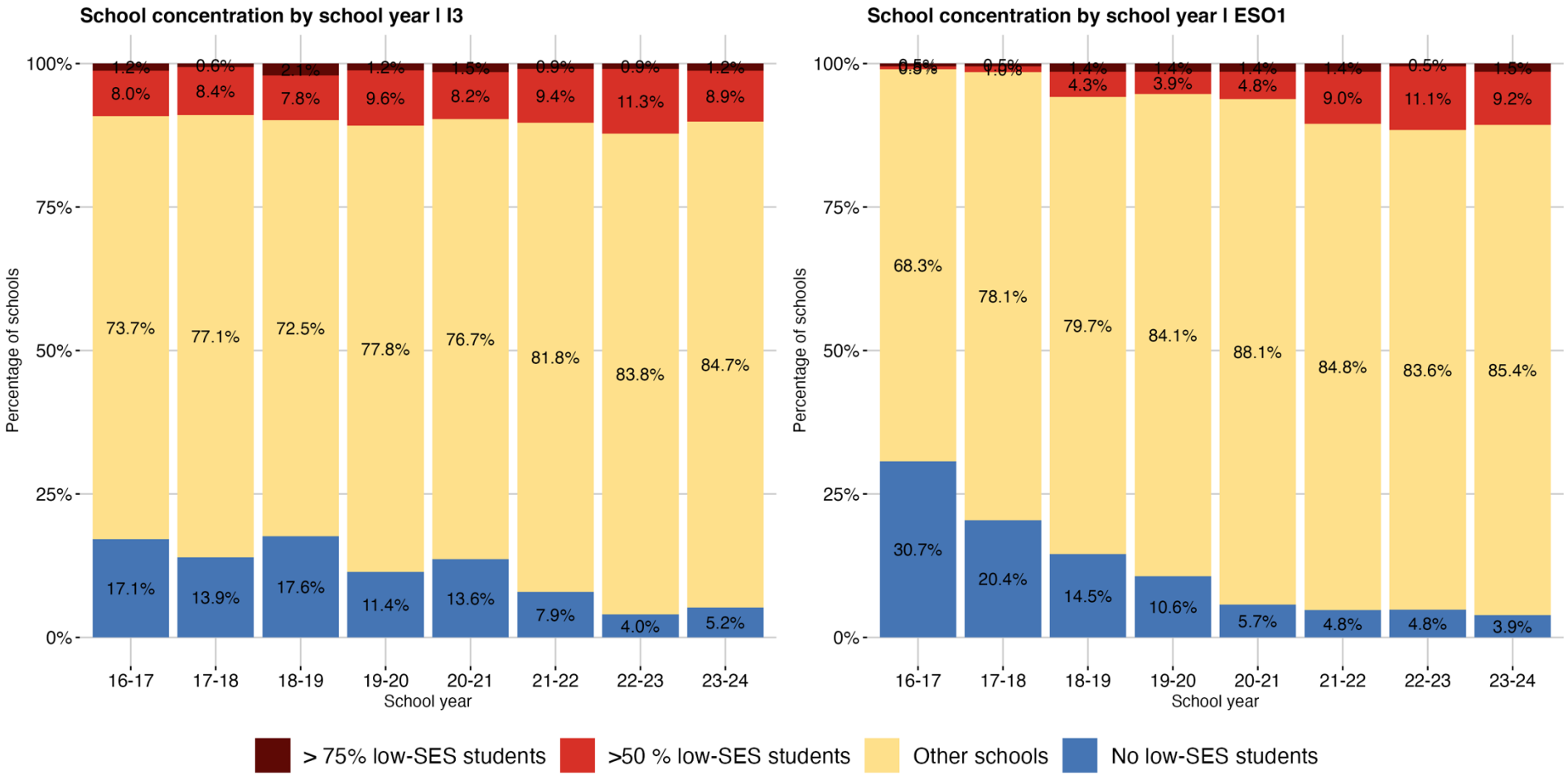
Understanding the impact: changes in patterns of segregation?





Extremely privileged | Very privileged | Privileged | Balanced | Ghettoised | Very ghettoised | Extremely ghettoised

Understanding the impact: different levels of concentration?



Discussion

No significant impact on segregation in the following years

Can the “maintance of the previous trend” be considered an impact of the Shock Plan?

Paradox? Reduction of the extremely segregated schools while the schools >50% of low-income students increase?

Increase of low-SES students: Composition effect?

Better targeting = awareness of concentration?

Catchment areas effects?

Next steps

Territorial analysis: catchment areas effects

Analysis on public/private schools

Focused analysis on migrant students
(first and second generation)